Project Documentation: BeeSmart Decision Support System

1. Executive Summary

The BeeSmart Decision Support System is a spatial visualization and predictive management platform designed to eliminate guesswork in apiary management. By integrating environmental data with real-time analytics, the system optimizes hive placement and maximizes honey production.

2. Problem Statement

Current beekeeping practices rely heavily on anecdotal experience and static seasonal patterns rather than data-driven insights. This traditional approach results in several critical inefficiencies:

- **Suboptimal Hive Placement:** Locations are chosen without precise data on nectar availability.
- **Reduced Honey Yield:** Inability to maximize the potential of specific forage areas.
- **Missed Flowering Windows:** Lack of alerts or tracking for rapid environmental changes.
- **Operational Risk:** High management overhead and risk in large-scale, multi-hive operations.

The Gap: There is currently no unified system that merges environmental data, predictive modeling, and high-fidelity spatial visualization to guide beekeeping decisions.

3. Target Audience (Personas)

3.1 Beekeepers (Field Users)

- **Role:** Hands-on management of individual colonies.
- **Primary Goals:** Monitoring hive health, choosing the best local spots for placement, and reacting to immediate local conditions.
- **Key Needs:** Mobile access, real-time environmental alerts, and easy data entry for inspections.

3.2 Apiary Admins / Farm Owners

- **Role:** Strategic oversight of multiple apiary sites and large-scale hive movements.
- **Primary Goals:** Tracking overall performance, managing logistics for hundreds of hives, and long-term strategic planning.

- **Key Needs:** Performance dashboards, regional suitability trends, and resource allocation tools.

4. System Workflow & User Experience

The application follows a "Map-First" philosophy, prioritizing visual data to drive decision-making.

Phase 1: Environmental Awareness (The Map)

The user begins by opening the interactive map.

- **Heatmap Visualization:** The system renders a dynamic suitability map.
- **Suitability Indicators:**
  - **Green:** Optimal regions (high flowering density, ideal weather).
  - **Yellow:** Moderate regions.
  - **Red:** Poor or declining regions.

Phase 2: Site Exploration

Users can interact with specific geographical points to dig deeper into the data.

- **Navigation:** Zooming and panning to find potential new sites.
- **Instant Analysis:** Tapping a location reveals specific environmental scores (e.g., humidity, floral biodiversity index, and wind protection).

Phase 3: Decision & Placement

- **Site Selection:** Based on the high-scoring regions identified in Phase 2, the user confirms a hive placement.
- **Deployment:** The hive is "virtually" placed on the map to begin tracking.

Phase 4: Active Management

- **Hive Tracking:** Users record data from physical inspections (queen health, pest levels).
- **Automated Alerts:** The system notifies the user if conditions at a current hive location deteriorate.

Phase 5: Predictive Planning

- **Forecasting:** Users utilize a "Time Slider" or prediction panel to view projected flowering conditions for the next 14–30 days.

- **Preparation:** Beekeepers prepare logistics for moving hives to the next high-yield zone before the window opens.

5. Key Success Metrics

- **Yield Increase:** Percentage increase in honey production per hive.
- **Operational Efficiency:** Reduction in time spent scouting for new locations.